

206 Quiz - July 2021

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The characteristic table of **AB flip-flop** is given below. If you are given a JK flip-flop to implement AB flip-flop, minimum how many additional basic gates will you have to use?

A	B	$Q(t+1)$
0	0	1
0	1	$Q'(t)$
1	0	$Q(t)$
1	1	0



operation gulon JK er songe same

- ☐ 2
- ☐ 3
- ☒ 1
- ☐ 4



If you want to implement a function $f(A,B,C,D)=\sum(0,1,4,6,7,8,10,12,14,15)$ using basic gates and a single 4-to-1 MUX with C & D as selector bits, then what would be equation for the input I_1 (i.e., the input line that would be selected when $CD=01$)?

- ☐ 1
- ☐ Correct answer not given
- ☐ AB
- ☐ $\sim A$
- ☒ $\sim(A+B)$

What is the minimum number of flipflops necessary to create a 4-bit Register that only holds the following states: 1, 3, 4, 8, 12 ?

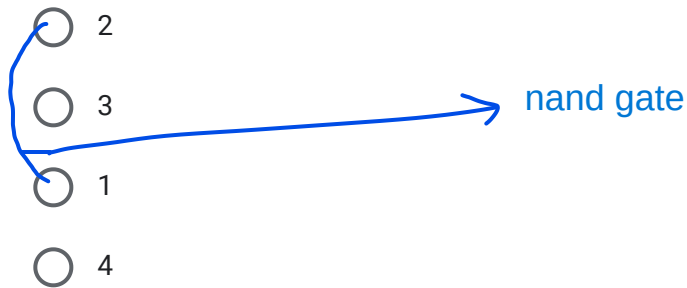
- ☐ 4
- ☐ 2
- ☒ 3
- ☐ 5

If $y'=(a(bcd)'e)'$, for $y=1$ the value of a, b, c, d, e will be-

- ☐ $a=b=c=d=e=0$
- ☐ $a=b=c=d=e=1$
- ☒ $a=e=1, b=c=d=0$
- ☐ $b=c=d=1, a=e=0$



You are given an edge triggered flip-flop without any **direct reset** input. All the gates used to design this flip-flop are 2-input. You are asked to add a direct reset input for this flip-flop. Minimum how many additional basic gates can you use?

- ☐ 2
 - ☐ 3
 - ☐ 1
 - ☐ 4
-  nand gate

The number of enables in 74138 is--

- ☐ 4
- ☒ 3
- ☐ 1
- ☐ 2

Which of the following is called equivalence gate--

- ☒ Xnor
- ☐ Nand
- ☐ Xor
- ☐ Nor



Gray code is used in K-Map because _____

- ☐ None of the other options
- ☐ in a code, there is odd number of 1s.
- ☐ in two adjacent codes, there is change in three bits.
- ☒ in two adjacent codes, there is change in one bit.

State whether each of the below statements are True or False. Putting a tick mark to the True statements will suffice.

- ☐ There is a minimum time called the setup time during which the D input of a D Flip-flop must not change after the application of the positive/negative transition of the clock
- ☒ T flip-flop can be constructed from JK flip-flop without using any additional gates
- ☒ The output of a latch cannot be applied through a combinational logic to the input of another latch.
- ☐ The most economical flip-flop is the edge triggered D flip-flop.



The simplified equation for the given K-map is -



		A_1A_0			
		00	01	11	10
A_3A_2	00	0	0	1	1
	01	1	1	0	0
	11	1	1	0	0
	10	0	0	1	1



- ☐ $Y = A_2A_3' + A_3A_2'$
- ☒ $Y = A_1' A_2 + A_2' A_1$
- ☐ $Y = A_1' A_3 + A_3' A_1$
- ☐ $Y = A_1' A_0 + A_0' A_1$

What is the minimum number of basic gates required to build a 4-to-1 MUX?
Assume, **2-input gates**.

- ☒ 13
- ☐ 7
- ☐ 9
- ☐ 5
- ☐ Correct answer not given



For 6 bit input, the maximum possible BCD value will be _____.

39

Your answer

$f(x,y,z) = (x \rightarrow y) \rightarrow \sim z$, select the **maxterms** for POS of $f(x,y,z)$ from 0, 1, 2, 3, 4, 5, 6, 7 (Select the checkboxes that apply)

- ☒ 1
- ☐ 2
- ☒ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☒ 7
- ☐ 0

Which of the following statements is False?

- ☒ You can simulate a mod 16 counter through a mod 8 counter and some combinatorial circuits.
- ☒ It is possible to pause a Counter without pausing the clock.
- ☐ All Counters are dependent on initial state.
- ☐ It is possible to transform one counter to another if both has same number of states.



Which one of the following is correct?

- ☐ None of the options
- ☐ For proper operation of 7483, its C_{in} must not be left open
- ☐ 7483 is a low speed 4 bit adder
- ☒ For proper operation of 7483, its C_{in} can be left open

The number of enables and type of enables **for each** 4x1 multiplexer module in 74153 is

- ☒ 1, Active low
- ☐ 2, Active high
- ☐ 2, Active low
- ☐ 1, Active high

Which of the following statements is True?

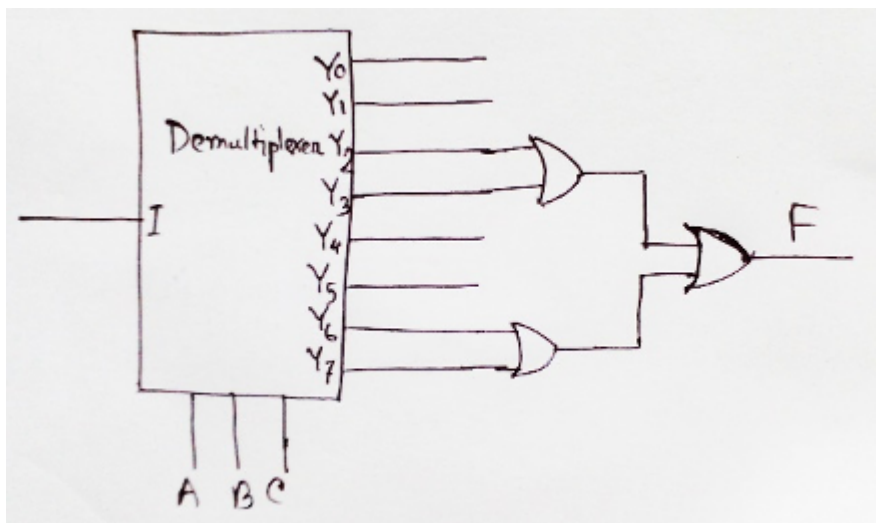
- ☐ You can design a Universal Shift Register with asynchronous flipflops.
- ☐ You cannot simulate Left/Right Shift in a register that only supports Parallel Load.
- ☐ It is impossible to simulate a SIPO Register through a PIPO Register.
- ☒ It is possible to design a Shift Register that takes 2 clock cycles to shift left by 1 bit.



What is the minimum number of flipflops required to output the following sequence: 0,1,0,2,0,1,0,2,...?

- ☐ 4
- ☐ Impossible
- ☐ 3
- ☒ 2

For the following Demultiplexer, if the Input $I=1$, then what will be the functional form of $F(A,B,C)$? Note that gates used in the figure are 2-input OR gates.



- ☐ AB
- ☐ A'B
- ☐ BC
- ☒ B
- ☐ Correct answer not given



Which one of the following is correct for Logisim?

- ☐ The circuits implemented in Logisim require Vcc and Gnd connection for proper operation.
- ☐ The circuits implemented in Logisim do not require Vcc and Gnd connection for proper operation
- ☐ The circuits implemented in Logisim require only Gnd connection for proper operation
- ☐ The circuits implemented in Logisim require only Vcc connection for proper operation.

What are the minimum number of two input NOR gates to perform XOR operation?

- ☐ 7
- ☐ 5
- ☒ 6
- ☐ 4

The SOP form for the function $f(A,B,C,D) = \sum m(0,2,4,8,10,14)$ is--

- ☐ $B'D + ACD' + A'C'D'$
- ☐ $B'D' + AC'D + A'C'D'$
- ☒ $B'D' + ACD' + A'C'D'$
- ☐ $BD' + ACD' + A'C'D'$



Excess 3 code for 12 will be --

- ☐ 0011
- ☒ 1111
- ☐ Dont Care
- ☐ 1100

You are provided with a D flip-flop and are asked to design a new flip-flop which incorporates **no change operation** along with the operations offered by the D flip-flop. **Minimum** how many additional basic gates do you need for accomplishing the task?

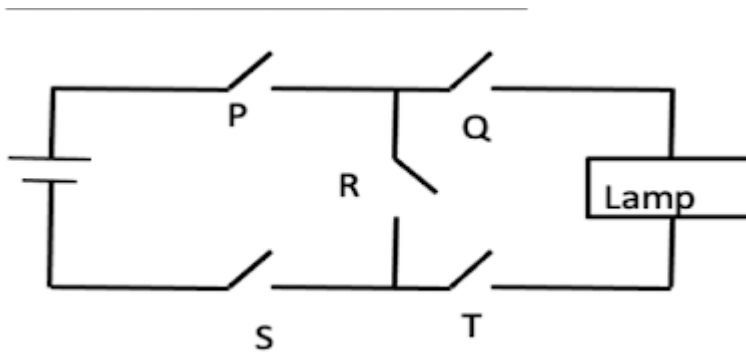
- ☐ 5
 - ☒ 3
 - ☐ 6
 - ☐ 2
- D to SR ff

It should be kept in mind that Don't Care terms should be used along with the terms that are present in-

- ☐ Expressions
- ☒ Minterms
- ☐ Latches
- ☐ K-Map



Select the logical function $f(P, Q, R, S, T)$ for the following circuit -



☐ $R(PS+QT)$

☒ $P(S(QT+R))$

☐ $PR+ST$

☐ $P(RS+QT)$

What is the minimum number of flipflops required to output the following sequence: 0,0,1,1,0,0,1,1,...?

☐ Impossible

☐ 1

☒ 2

☐ 3



A three (3) variable K-map contains _____ cells.

- ☐ 4
- ☐ 32
- ☒ 8
- ☐ 16

Four routers are connected in a network that can be represented by the logical function $f(w, x, y, z) = wx' + y(z+x) + wz' + x'$. For AND gate there is a delay of 5ns, and for OR gate the delay is 6ns. What is the value of minimum possible delay of that function f in ns? (write only the value. For example: 100)

39

Your answer

The POS form for the function $f(A,B,C,D) = \sum m(0,2,4,8,10,14)$ is--

- ☒ $D'(A+B'+C')(A'+B'+C)$
- ☐ $D'(A+B'+C)(A'+B'+C)$
- ☐ $D'(A+B'+C')(A+B'+C)$
- ☐ $D'(A+B'+C')(A'+B+C)$



A boolean function with five (5) variables has an octet of one (1). How many variable(s) will the final literal contain?

- ☐ 3
- ☐ 1
- ☐ 4
- ☒ 2

What is the minimum number of flipflops required to produce the output sequence of the 4-bit Johnson counter?

- ☐ 3
- ☒ 4
- ☐ 2
- ☐ 5

Which of the following is NOT considered for forming groups in K-map?

- ☐ Vertical
- ☐ Rolling
- ☐ Horizontal
- ☒ Diagonal



The minimized form of $f(A,B,C)=AB + A'C+BC$ is--

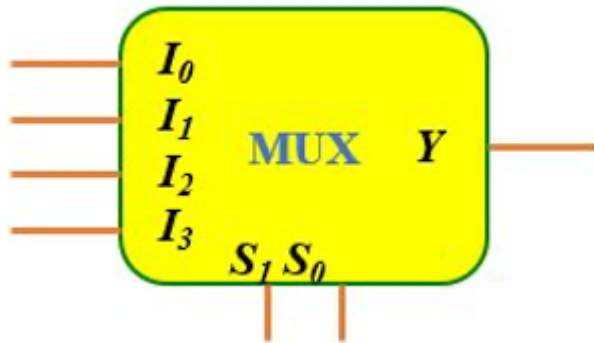
- ☐ $B(A+C)$
- ☒ $AB + A'C$
- ☐ $C(A'+B)$
- ☐ None of the above

The outputs in 74138 are--

- ☐ active low and active high
- ☐ active high
- ☐ None of the options
- ☒ active low



Consider the following 4-to-1 MUX. Suppose, the S_0 selector bit of the MUX is malfunctioning and so, you want to use it as a 2-to-1 MUX. How many additional gates are required, at minimum, to make it usable as a 2-to-1 Mux?



- ☐ 2
- ☐ It is not possible to use this as a 2-to-1 MUX
- ☐ 1
- ☐ 4
- ☒ 0

You have a 4-bit SIPO Shift Register (i.e. NO PARALLEL LOAD) that currently holds 1001. What is the minimum number of clock cycles required for the shift register to hold 1000?

- ☐ 4
- ☐ 2
- ☒ 3
- ☐ 1



Which one of the following is correct?

- ☐ 7483 has no internal carry look ahead
- ☐ 7483 is cascadable in 3 bit increments
- ☒ 7483 is cascadable in 4 bit increments
- ☐ 7483 is cascadable in 2 bit increments

$P \oplus Q = R$, then

- ☐ $P \oplus R = R'$
- ☐ $Q \oplus R = PQ$
- ☐ Option 5
- ☒ $P \oplus R = Q$
- ☐ $P \oplus R = Q'$

Can you use T-flipflops to create a Universal Shift Register?

- ☐ No
- ☒ Yes

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